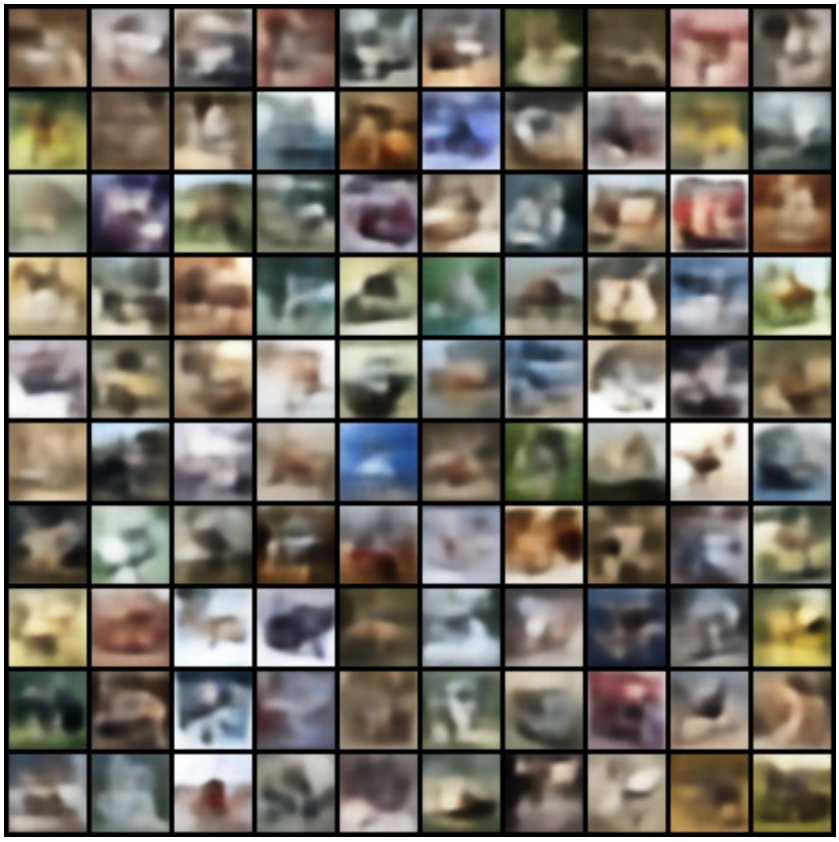
Assignment 2 — Image Generation using (beta-)VAE, VQ-VAE and WGAN

Course: Unsupervised Deep Learning (S1-26_AIMLZG533) Assessment: Assignment 2 Group: 91

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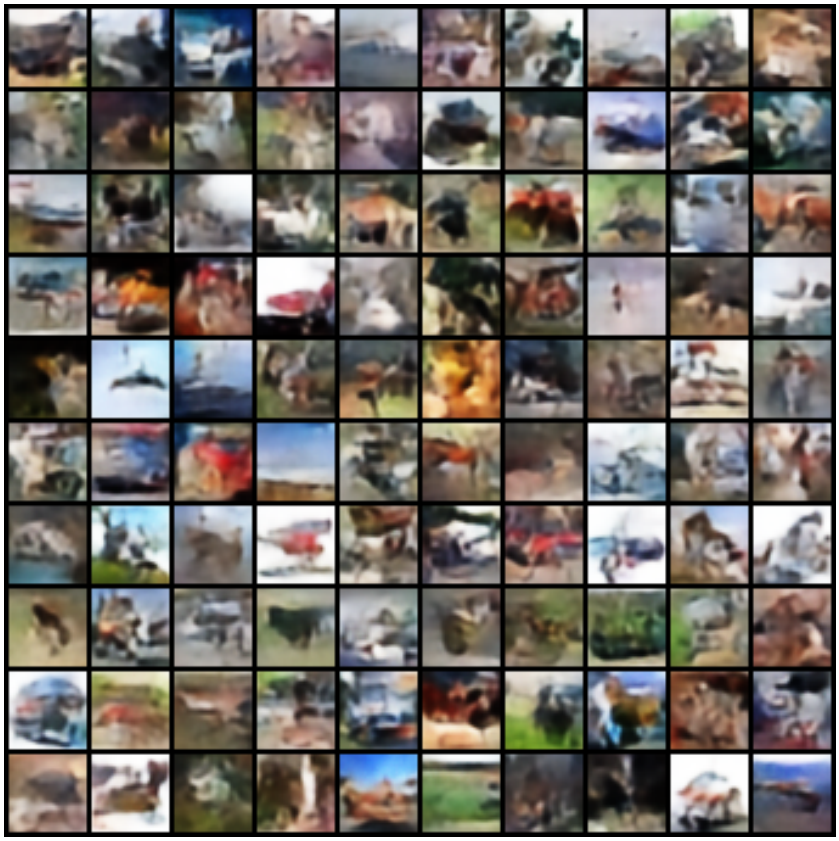
100 Visual Examples per Model

VAE (beta=1)



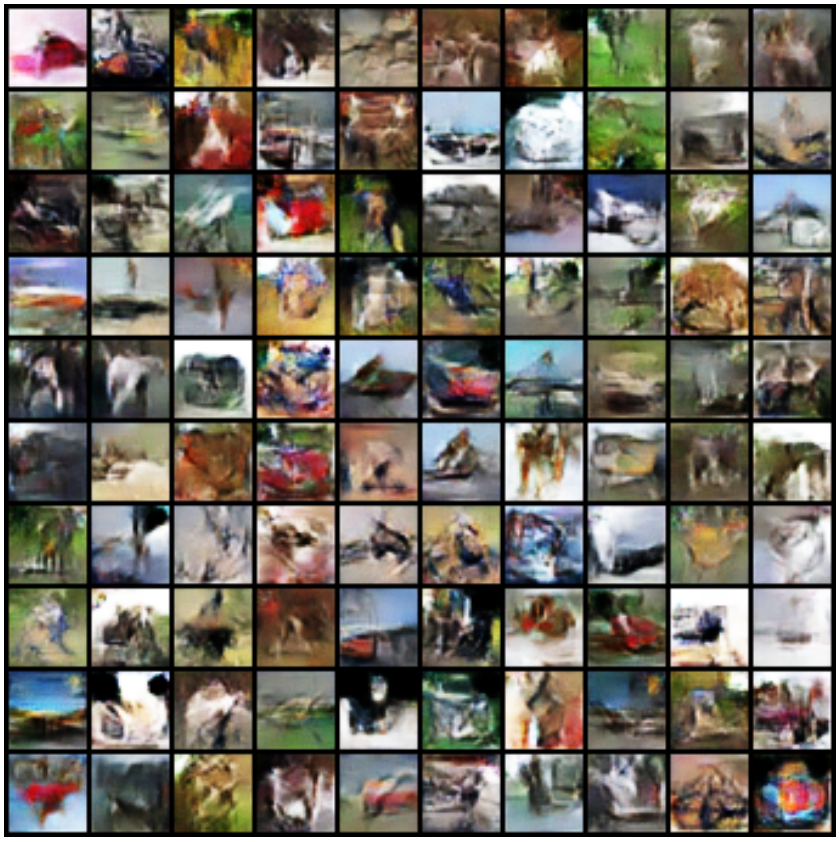
Edge-sharpness proxy (Laplacian variance) = 4767.3; colour variation (RGB std) = 64.3; tile-to-tile diversity = 46.5. High Laplacian variance indicates sharp edges and fine detail. Tile-to-tile diversity suggests varied content across the 100 samples.

VQ-VAE (K=256)



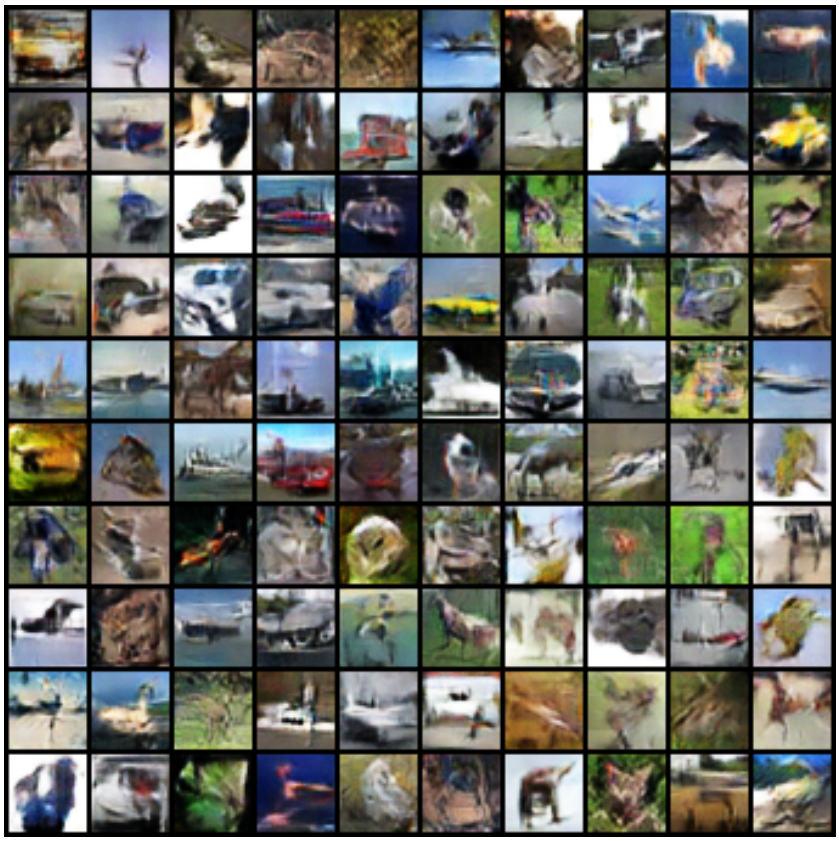
Edge-sharpness proxy (Laplacian variance) = 5558.8; colour variation (RGB std) = 69.6; tile-to-tile diversity = 52.4. High Laplacian variance indicates sharp edges and fine detail. Tile-to-tile diversity suggests varied content across the 100 samples.

WGAN



Edge-sharpness proxy (Laplacian variance) = 6771.7; colour variation (RGB std) = 71.3; tile-to-tile diversity = 55.7. High Laplacian variance indicates sharp edges and fine detail. Tile-to-tile diversity suggests varied content across the 100 samples.

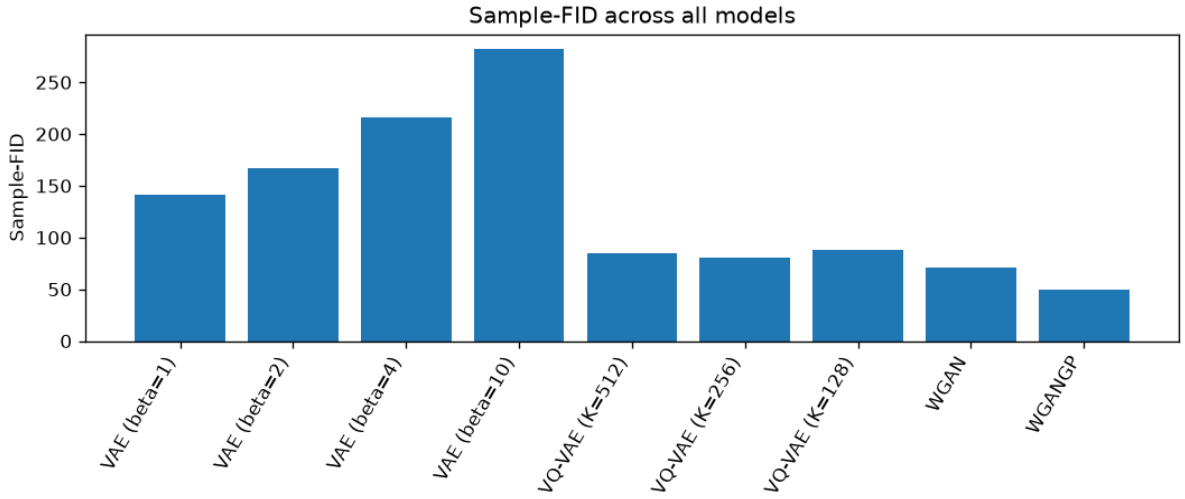
WGAN-GP



Edge-sharpness proxy (Laplacian variance) = 6404.9; colour variation (RGB std) = 71.0; tile-to-tile diversity = 55.6. High Laplacian variance indicates sharp edges and fine detail. Tile-to-tile diversity suggests varied content across the 100 samples.

Evaluation Score Comparison

Model	Test PSNR (dB)	Recon-FID	Sample-FID	Train time (s)	Sample time /10k (s)	Sample time /100 (s)
VAE (beta=1)	19.71	126.51	141.20	227	0.00	0.0000
VAE (beta=2)	18.49	155.04	167.09	214	0.00	0.0000
VAE (beta=4)	17.32	207.77	215.98	243	0.00	0.0000
VAE (beta=10)	16.02	276.87	282.45	255	0.00	0.0000
VQ-VAE (K=512)	25.43	49.89	84.53	827	36.85	0.3685
VQ-VAE (K=256)	24.90	53.88	81.03	788	36.66	0.3666
VQ-VAE (K=128)	24.29	62.19	87.69	1016	36.53	0.3653
WGAN	N/A - no encoder	N/A - no encoder	70.69	2791	0.03	0.0003
WGANGP	N/A - no encoder	N/A - no encoder	49.84	5788	0.01	0.0001



Comparative Discussion

Reconstruction error

The best (beta-)VAE reconstruction PSNR was 19.71 dB at beta=1, versus 25.43 dB for VQ-VAE at K=512. GANs have no encoder and so no reconstruction PSNR is reported (N/A). The well-established expectation is that VQ-VAE's discrete, higher-capacity codebook gives sharper, higher-PSNR reconstructions than a continuous-latent (beta-)VAE, especially as beta grows and the VAE is pushed toward blurrier outputs by the KL term; our measured numbers agree with that expectation.

Sampling quality

The lowest (best) sample-FID overall was 49.84, achieved by WGANGP. GANs are expected to achieve the sharpest samples and the best FID among the three families because they optimise sample realism directly (an adversarial loss) rather than a pixel-wise or likelihood-based objective; VAEs are expected to produce visibly blurrier samples due to the Gaussian-decoder/pixel-wise reconstruction loss; VQ-VAE sample quality is bounded by how well the separately-trained PixelCNN prior models the discrete latent distribution, and PixelCNN sampling is sequential (one raster position at a time) and therefore much slower than a single forward pass through a VAE decoder or GAN generator.

Speed

PixelCNN sampling of 10,000 latent maps took [36.9, 36.7, 36.5] s across K=[512, 256, 128] (sequential, one 8x8 position at a time), versus [0.03, 0.01] s for 10,000 WGAN/WGAN-GP samples (a single forward pass per batch). This matches the expected ordering: autoregressive priors are far slower to sample from than single-forward-pass generators/decoders.

Experimental Setup

VAE: betas=[1, 2, 4, 10], batch size 128, Adam lr=1e-3. VQ-VAE: K=[512, 256, 128], batch size 128, Adam lr=2e-4, EMA decay 0.99, commitment 0.25. PixelCNN prior: 12 gated residual layers, 128 channels, Adam lr=3e-4, batch 128. WGAN: RMSprop lr=5e-5, weight clip [-0.01,0.01], n_critic=5. WGAN-GP: Adam lr=1e-4 (betas 0,0.9), gradient penalty lambda=10, n_critic=5. FID protocol: torchmetrics FrechetInceptionDistance(feature=2048, normalize=True), 10,000 real CIFAR-10 test images vs 10,000 generated images for every model (same protocol throughout, so the comparison across models is fair even though this is smaller than the 50,000-image convention used in some papers).